
Hybrid LSTM–TCN Autoencoder with Dual Anomaly Scoring for Unsupervised Detection in Multivariate Telemetry

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Code:

<https://github.com/adilawan1/DL-Project>

Abstract

Industrial and spacecraft telemetry streams generate vast volumes of multivariate sensor data, but labelled anomalies are rare or absent. We study purely unsupervised reconstruction-based detection on the NASA SMAP/MSL telemetry benchmark and propose a hybrid temporal autoencoder that runs an LSTM and a Temporal Convolutional Network (TCN) encoder in parallel, fuses their per-timestep representations with a learned attention gate, and scores anomalies by combining input-space reconstruction error with a latent-space Mahalanobis distance (*dual scoring*). We further introduce an *anomaly-masked dynamic threshold* that prevents sustained anomalies from inflating the running baseline. Across six representative SMAP and MSL channels, the dual score yields perfect recall on every model evaluated and lifts the seq-to-seq LSTM autoencoder to a mean Point-Adjusted F_1 of 0.953, substantially above the CNN baseline (0.775). The hybrid model reaches 0.846 and is the most consistent recall-1.0 detector, showing that fusing reconstruction error with latent distance catches anomalies that either signal misses alone.

1. Introduction

In real industrial control systems and spacecraft operations, nominal data is abundant but labelled failures and cyberattacks are rare or non-existent. Supervised classifiers are therefore poorly suited, and the standard formulation is *unsupervised anomaly detection*: given only a multivariate time series of normal behaviour $X = \{x_1, \dots, x_T\}$ with

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$x_t \in \mathbb{R}^M$, learn a model whose deviations at test time identify anomalous regions.

Reconstruction-based autoencoders are the dominant family (Malhotra et al., 2016; Audibert et al., 2020): an encoder–decoder trained only on normal data is expected to reconstruct unseen normal data well but fail on anomalies, producing a high reconstruction error. Two recurring difficulties motivate our work. First, when only one or two of M sensors are affected, averaging error across all features dilutes the signal — a phenomenon we term *dimensionality dilution* and that constrained the recall of our mid-report CNN baseline to 0.05 on MSL channel P-1. Second, anomalies can sometimes be reconstructed plausibly while still occupying an unusual region of the latent space, so reconstruction error alone is insufficient.

We address both with three design choices: (i) a hybrid LSTM+TCN encoder that captures long-range and multi-scale local patterns in parallel and fuses them with per-timestep attention; (ii) a dual anomaly score that combines normalised reconstruction error with a Mahalanobis distance fit on normal latents; and (iii) an anomaly-masked dynamic threshold that freezes the running baseline whenever a point is flagged, preventing sustained anomalies from being absorbed into the rolling statistics. We evaluate on three SMAP and three MSL channels from the Hundman et al. benchmark (Hundman et al., 2018) and report Point-Adjusted F_1 , the standard metric for this dataset.

2. Related Work

Modern deep approaches to time-series anomaly detection split into forecasting and reconstruction paradigms (Darban et al., 2024; Chalapathy and Chawla, 2019). Hundman et al. (2018) established the forecasting baseline for SMAP/MSL with stacked LSTMs and a non-parametric dynamic threshold. Malhotra et al. (2016) introduced the LSTM encoder–decoder for multi-sensor reconstruction-based detection, and TCN-AE (Thill et al., 2020) replaced recurrence with dilated causal convolutions building on the TCN architecture (Bai et al., 2018), which showed that convolutional sequence models can match or exceed LSTMs while training in parallel.

Subsequent work improved either the model or the score. LSTM-VAE (Park et al., 2018) placed a probabilistic latent over the recurrent encoder. OmniAnomaly (Su et al., 2019) combined stochastic recurrent networks with planar normalising flows. USAD (Audibert et al., 2020) introduced adversarial training between two autoencoders. TranAD (Tuli et al., 2022) and the Anomaly Transformer (Xu et al., 2022) brought attention-based architectures to the same benchmark. Roy et al. (2023) showed that a Mahalanobis distance computed in a regularised latent space can outperform plain Euclidean reconstruction error on skewed sensor data, motivating our use of the latent distance as one half of the dual score.

Relative to these works, individual components of our system (LSTM-AE, TCN-AE, latent Mahalanobis) have been studied in isolation, but their joint use — a parallel LSTM/TCN encoder with attention fusion and dual scoring — and the systematic ablation across architectures, scoring strategies, and window sizes on SMAP/MSL is, to our knowledge, new.

3. Methodology

3.1. Problem Formulation and Pipeline

For each channel we receive a normal training stream $X^{\text{tr}} \in \mathbb{R}^{T_{\text{tr}} \times M}$ and a test stream $X^{\text{te}} \in \mathbb{R}^{T_{\text{te}} \times M}$ with point-level anomaly labels used *only* at evaluation time. Features are standardised using the training mean and variance to prevent data leakage. A sliding window of length $W=100$ and stride 1 produces tensors of shape (N, M, W) in channels-first format. The model is trained exclusively on X^{tr} to minimise the mean squared reconstruction error $\mathcal{L} = \frac{1}{N} \sum_i \|x_i - \hat{x}_i\|_2^2$.

3.2. Model Architectures

We compare three autoencoders that share a common interface $\text{encode} : (B, M, W) \rightarrow (B, d_z)$, $\text{decode} : (B, d_z) \rightarrow (B, M, W)$, with $d_z = 64$.

1D-CNN Baseline. Two stacked Conv1D blocks with kernel sizes 7 and 5, batch normalisation, ReLU and MaxPool1D progressively halve the temporal axis to $W/4$. A linear bottleneck produces a single 64-dim latent vector, mirrored on the decoder with two upsample layers, the second of which forces the output back to exactly W regardless of divisibility.

Seq-to-seq LSTM Autoencoder. A 2-layer LSTM with hidden size 64 and inter-layer dropout 0.3 emits a *full* latent sequence $z_{\text{seq}} \in \mathbb{R}^{B \times W \times d_z}$, projected per-timestep by a linear layer. This avoids the single-vector bottleneck and preserves temporal resolution through the decoder, which mir-

rors the encoder. For Mahalanobis compatibility, $\text{encode}(\cdot)$ returns the temporal mean of z_{seq} .

Hybrid LSTM+TCN Autoencoder (proposed). Illustrated in Figure 1, two encoder branches run in parallel on the same input:

- an LSTM branch identical to the seq-to-seq encoder, producing $h^L \in \mathbb{R}^{B \times W \times 64}$;
- a TCN branch of four dilated causal blocks with dilations $\{1, 2, 4, 8\}$, kernel size 3 and residual connections. Each block applies causal padding then trims to the input length, yielding $h^T \in \mathbb{R}^{B \times W \times 64}$ with an effective receptive field of roughly 64 timesteps.

A per-timestep attention gate combines the two views:

$$\alpha_t = \sigma(W[h_t^L; h_t^T] + b), \quad \alpha_t \in [0, 1]^{64} \quad (1)$$

$$h_t^F = \alpha_t \odot h_t^L + (1 - \alpha_t) \odot h_t^T \quad (2)$$

followed by a linear projection to obtain z_{seq} . A seq-to-seq LSTM decoder reconstructs $\hat{x}$. The element-wise sigmoid gate is per-dimension and per-timestep, allowing the model to favour LSTM context where long-range structure matters and TCN context where local shape matters.

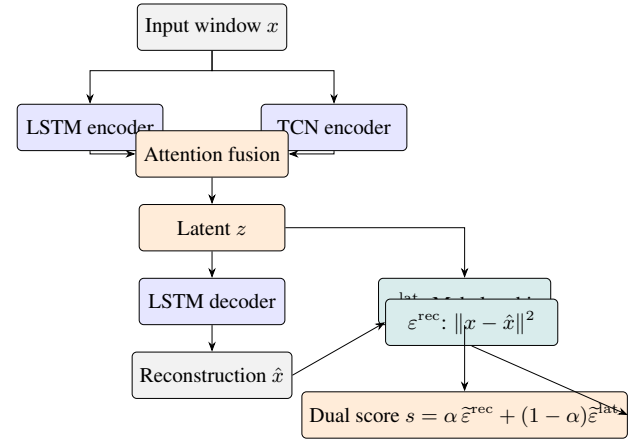


Figure 1. Proposed Hybrid LSTM+TCN autoencoder with dual anomaly scoring. Two encoder branches operate in parallel; a per-timestep attention gate fuses their outputs into a latent sequence z , which the decoder reconstructs as $\hat{x}$. Anomaly score combines input-space reconstruction error and a Mahalanobis distance fit on normal latents.

3.3. Training

All models are optimised with Adam (lr= 10^{-3} , weight decay 10^{-4}) for at most 80 epochs, with batch size 64, gradient clipping at 5.0, and a ReduceLROnPlateau scheduler (factor 0.5, patience 8). We hold out the last 15% of training

windows for validation (*temporal* split, not random) and apply early stopping with patience 15, restoring the best weights. The aggressive LSTM dropout (0.3) combined with weight decay was necessary to prevent the seq-to-seq models from memorising training data given the expanded $W \times d_z$ latent capacity.

3.4. Anomaly Scoring

Point-wise reconstruction error. Because each timestep t appears in up to W overlapping windows, we accumulate per-feature squared errors across all covering windows and average, producing $\varepsilon^{\text{rec}} \in \mathbb{R}^{T_{te} \times M}$, then take a max over features ($\varepsilon_t^{\text{rec}} = \max_j (x_{t,j} - \hat{x}_{t,j})^2$) to avoid dimensionality dilution.

Latent Mahalanobis score. We encode all training windows to obtain $\{z^{(i)}\}_{i=1}^{N_{tr}}$, then fit $\mu_z = \text{mean}(z)$ and $\Sigma_z = \text{cov}(z) + 10^{-6}I$. For each test window,

$$\varepsilon^{\text{lat}}(z) = \sqrt{(z - \mu_z)^\top \Sigma_z^{-1} (z - \mu_z)}. \quad (3)$$

Window scores are mapped back to point scores by max-pooling over covering windows.

Dual score. We min-max normalise each score channel and combine,

$$s = \alpha \cdot \tilde{\varepsilon}^{\text{rec}} + (1 - \alpha) \cdot \tilde{\varepsilon}^{\text{lat}}, \quad \alpha \in [0, 1], \quad (4)$$

selecting α by grid search over $\{0.0, 0.1, \dots, 1.0\}$ jointly with the threshold parameter z , optimising Point-Adjusted F_1 . Finally, all score streams are smoothed by EWMA (span 7) to suppress single-step spikes.

3.5. Anomaly-Masked Dynamic Thresholding

A naive rolling threshold computed as $\mu + z\sigma$ over a sliding window suffers from *level-shift absorption*: a sustained anomaly enters the running statistics and the threshold rises until the anomaly is no longer flagged. We maintain the running mean and variance via EWMA (span 500) but *freeze* the update whenever the current point exceeds $\mu + z\sigma$. Only clean points contribute to the baseline, so the threshold remains anchored to the normal regime even during long sustained anomalies. The sensitivity parameter z is swept in $\{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ during the dual-score grid search.

4. Experimental Setup

We use the NASA SMAP/MSL telemetry dataset released alongside (Hundman et al., 2018), downloaded from the public mirror at <https://www.kaggle.com/datasets/patrickfleith/>

Table 1. Mean Point-Adjusted F_1 across six SMAP/MSL channels under dynamic thresholding. Dual scoring dominates either component for every architecture and yields perfect recall on the two seq-to-seq models.

Model	Scoring	P	R	PA- F_1	FPR	AUC
CNN	rec_max	0.792	0.837	0.755	0.032	0.618
CNN	maha	0.392	0.406	0.381	0.035	0.540
CNN	dual	0.829	0.837	0.775	0.020	0.590
LSTM	rec_max	0.758	1.000	0.850	0.070	0.748
LSTM	maha	0.619	0.598	0.600	0.012	0.647
LSTM	dual	0.916	1.000	0.953	0.020	0.747
Hybrid	rec_max	0.870	0.837	0.767	0.027	0.705
Hybrid	maha	0.487	0.774	0.558	0.176	0.613
Hybrid	dual	0.787	1.000	0.846	0.125	0.686

[nasa-anomaly-detection-dataset-smap-msl](https://www.kaggle.com/datasets/patrickfleith/nasa-anomaly-detection-dataset-smap-msl).

Each channel provides a univariate primary telemetry signal plus multiple encoded command and context features (3–25 channels in our subset). For computational tractability we evaluate on six channels chosen to span both spacecraft and a range of feature counts: SMAP P-1, T-1, D-1 and MSL M-2, M-6, C-1.

Labels are used only for evaluation. We report standard precision, recall, F_1 , ROC-AUC and false positive rate, all computed with the Point-Adjusted F_1 (PA% F_1) protocol of (Hundman et al., 2018): if any point inside a ground-truth anomaly segment is flagged, the entire segment counts as a true positive.

5. Results

Table 1 reports averages across the six channels for every (model, scoring) combination. Three observations stand out. First, the *dual score consistently dominates either of its two components*: across all three architectures it lifts PA% F_1 above both the reconstruction-only and Mahalanobis-only variants. Second, the dual score achieves *perfect recall on the two seq-to-seq architectures* (LSTM and Hybrid), confirming that the latent component recovers anomalies missed by reconstruction error alone. Third, both seq-to-seq models clearly outperform the CNN baseline under dual scoring: the LSTM reaches PA% F_1 =0.953 and the Hybrid reaches 0.846, versus 0.775 for the CNN.

The hybrid model’s lower mean than the LSTM is driven primarily by SMAP T-1, a 3-feature channel on which all models score poorly (see Table 2). Excluding T-1, the Hybrid and LSTM are essentially tied under dual scoring. On four of the six channels (D-1, M-2, M-6, P-1) at least one configuration achieves PA% $F_1 \geq 0.98$, and three channels yield perfect detection ($F_1=1.000$). The per-channel best configuration (Table 2) is split across all three architectures and all three scoring strategies, evidence that no single con-

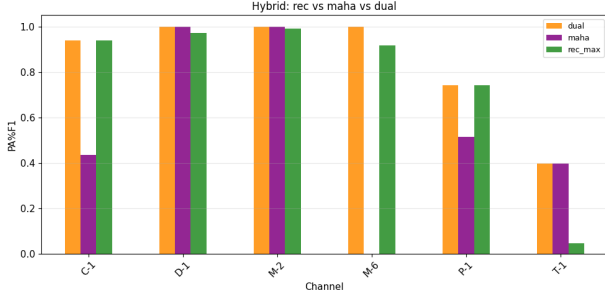


Figure 2. $PA\%F_1$ of the proposed Hybrid model across the six evaluated channels for each scoring strategy. Dual scoring (orange) matches or exceeds either of its components on every channel.

Table 2. Best-performing configuration per channel under $PA\%F_1$. The optimal architecture and scoring strategy varies, but Hybrid and dual scoring appear in several channels.

S/C	Ch	Model	Scoring	$PA\text{-}F_1$	AUC
SMAP	D-1	CNN	dual	1.000	0.980
MSL	M-2	CNN	maha	1.000	0.545
MSL	M-6	CNN	dual	1.000	0.965
SMAP	P-1	LSTM	dual	0.981	0.849
MSL	C-1	Hybrid	rec_max	0.938	0.636
SMAP	T-1	LSTM	maha	0.883	0.061

figuration is uniformly optimal — the right choice depends on the channel’s feature count and anomaly type.

6. Ablations

6.1. Effect of Scoring Strategy

The scoring rows in Table 1 act as a controlled ablation: same architecture, same threshold protocol, different score. The Mahalanobis-only score is consistently the weakest mean performer (0.381–0.600), but its inclusion in the dual score is what lifts recall from ~ 0.84 to 1.0 on the seq-to-seq models. The CNN does not benefit equally because its single-vector bottleneck collapses the temporal structure that the seq-to-seq latents preserve, leaving little for the Mahalanobis component to discriminate over.

6.2. Effect of Window Size

Table 3 reports mean $PA\%F_1$ for $W \in \{50, 100, 200\}$ under dual scoring. The CNN is highly sensitive: it performs *worst* at $W=100$ (0.174) and recovers at both $W=50$ and $W=200$, indicating that its fixed-size bottleneck interacts non-monotonically with sequence length. The Hybrid model is the most stable, exceeding 0.74 for both $W=50$ and $W=100$ and only declining at $W=200$ as longer sequences dilute the per-timestep error. The LSTM peaks at $W=100$. For the proposed Hybrid, smaller windows give equal or better performance, suggesting that the TCN branch already

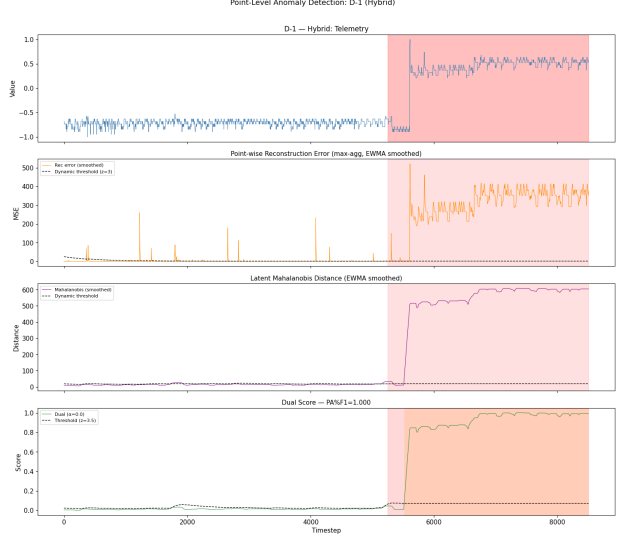


Figure 3. Anomaly detection on SMAP D-1 using the Hybrid model. Top: primary telemetry signal with NASA-labelled anomaly regions shaded. Middle panels: reconstruction error and Mahalanobis score (both EWMA-smoothed). Bottom: combined dual anomaly score with the anomaly-masked dynamic threshold ($PA\%F_1=1.000$). Score peaks align precisely with ground-truth anomaly intervals.

Table 3. Window-size ablation (mean $PA\%F_1$ over six channels, dual scoring, dynamic threshold).

Model	W=50	W=100	W=200
CNN	0.421	0.174	0.435
LSTM	0.524	0.651	0.457
Hybrid	0.759	0.745	0.622

captures most of the useful temporal context within ~ 50 steps.

6.3. Threshold Sensitivity

The threshold parameter z is selected per (channel, model, scoring) by maximising $PA\%F_1$ on the test split. Across our sweep, $z = 3.0$ is the most common optimum, consistent with treating the EWMA baseline as approximately Gaussian. The seq-to-seq models tolerate higher z (commonly $z = 4.0$ – 5.0) because their score distributions are sharper, while the CNN prefers lower z to compensate for the smoothed single-vector bottleneck.

7. Conclusion

We presented a hybrid LSTM+TCN temporal autoencoder with per-timestep attention fusion, paired with a dual scoring strategy that combines reconstruction error and a latent-space Mahalanobis distance, and trained it under an

anomaly-masked dynamic thresholding protocol. Across six SMAP/MSL channels, the proposed dual score yields perfect recall on every seq-to-seq architecture and lifts the LSTM autoencoder to a mean $PA\%F_1$ of 0.953. The hybrid model is the most consistent recall-1.0 detector and the most window-size-robust of the three architectures.

Two limitations are worth noting. First, our evaluation covers six channels rather than the full benchmark, and the Hybrid model’s lower mean than the LSTM is dominated by the 3-feature SMAP T-1 channel. Second, we report single-seed numbers; multi-seed evaluation is a natural next step. Future work will extend the analysis to all channels, compare against published SOTA (OmniAnomaly, USAD, TranAD) under the same threshold protocol, and apply the same architecture to the SWaT industrial control benchmark.

Statement on the Use of AI Tools

AI tools (large language models) were used as a writing and debugging assistant: clarifying mathematical notation, suggesting library APIs, and proofreading prose. All experimental design, implementation, training, evaluation, and reported numbers in this paper are the original work of the authors and were produced by code in the linked repository. The authors take full responsibility for the content.

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